

BEHAVIOR contact-rich needs wrist force.

Keep the modern VLA/FM controller. Make calibrated 6-axis F/T a first-class late token with sparse expert fusion so insert/wipe/occlusion phases stop being pure vision failures.

NO Discord / Drive

No invented BEHAVIOR SR

No $\pi_{0.5}$ bake-off numbers

ForceVLA = 2025 blueprint

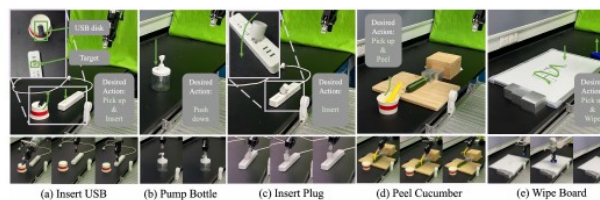


Figure 4: Overview of task setups used in evaluation. (a) Insert USB, (b) pump bottle, (c) insert plug, (d) peel cucumber, and (e) wipe board. These tasks span diverse contact dynamics and manipulation skills, from precise insertions to tool-mediated surface interactions.

5 Experiments

This section presents a comprehensive suite of real-world contact-rich manipulation experiments and analytical studies to empirically validate the ForceVLA model. The evaluation is structured around four core research questions: (1) the overall effectiveness of ForceVLA compared to baselines that

Five facts that do not change with the seat

Late 6-axis F/T + FVLMoE

01

After VLM (PaliGemma/SigLIP): project wrench $\rightarrow$ force token $\rightarrow$ FVLMoE (E=4, top-1) $\rightarrow$ additive guide into π_0 flow-matching action head. Force stays late.

Contact-rich on Flexiv

02

Insert USB / plug, pump bottle, wipe board, peel cucumber. ForceVLA-Data: 244 traj / ~140k steps / ~50 demos per task. Quest3 teleop.

Must-cites

03

Avg 60.5 vs π_0 -base w/o F 37.3 (+23.2). Naive w/ F only 40.2 (+2.9). Early MoE-before-VLM 0% $\rightarrow$ late FVLMoE 80% (Tab.3). Occlusion 90%. FAST+naive force 31.0 $\rightarrow$ 14.2.

CLEAR contrasts

04

NOT MoPA-PD ancestor (MoPA = MP privilege $\rightarrow$ vision student; ForceVLA = deploy force into VLA). NOT JEPA/WM child. No $\pi_{0.5}$ bake-off — do not invent.

Number hygiene

05

Typo fix: 37.3 = w/o F (not w/ F). Preserve paper numbers. No invented BEHAVIOR SR. Pump caveat: naive w/F 83 > ForceVLA 67 — do not overclaim.

Household activities are dense with contact phases

Insert / seat

Plugs, USB, chargers, appliance connectors. Misalignment + occlusion — Fig.1 failure mode.

Drawers / lids / cabinets

Contact detection + compliant push/pull. Expert routing for free-space vs contact phases.

Wipe / clean / spill

Continuous force maintenance. Wipe Board and peel analogs in ForceVLA-Data.

Crowded counters

Wrist cam occludes contact. Lean on force when vision degrades — Tab.2 occlusion 90% moral.

Food prep force

Peel/cut/press phases. Continuous contact metrics (length $\uparrow$ / strokes $\downarrow$), not only binary SR.

Open-world $\neq$ contact-solved

π_0 / $\pi_{0.5}$ semantic strength does not erase wrench-dominated failure modes.

BEHAVIOR need → ForceVLA-shaped response

Plug / peg / USB / charger

F/T-conditioned insert chunks; occlusion robustness (90% moral)

Drawers / dishwashers / lids

Contact detection + compliant phases via expert routing

Wiping / cleaning surfaces

Continuous force maintenance (wipe/peel analogs)

Crowded counters / clutter

Lean on force when wrist cam occludes contact

Hybrid with $\pi_{0.5}$ / RTC stacks

TENTATIVE: HL semantic loop + low-level FM+FVLMoE when wrist F/T present — NOT claimed by paper

Hardware path

Arms with integrated or retrofit F/T; calibrate; paper's estimated-wrench caveat

Drop-in late MoE on OpenPI-style FM stacks

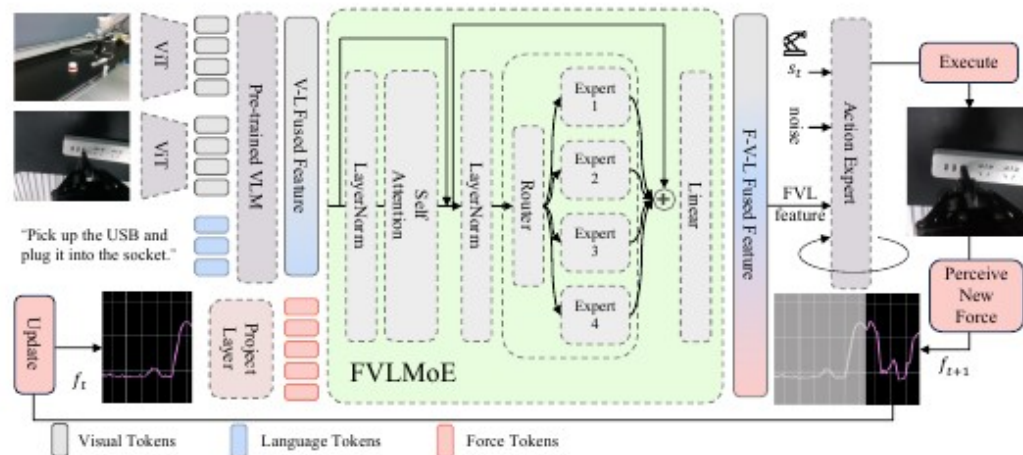


Figure 3: Overview of our ForceVLA model. Visual and language inputs are processed by a pre-trained VLM to form contextual embeddings. External force signals are projected and fused with VLM outputs via the FVLMoE module. The resulting multimodal features guide a flow-based action head to generate contact-aware robot actions.

that are used to select a small subset of k experts (typically $k = 1$ or $k = 2$, where $k \ll N$) from the total pool of N experts. The input token x is then routed only to these k active experts. The outputs of these active experts, $E_i(x)$, are subsequently aggregated, commonly through a weighted sum where the weights $g_i(x)$ are also derived from the gating network. The final output $y(x)$ of the MoE layer can be expressed as: $y(x) = \sum_{i \in \text{TopK}(G(x))} g_i(x) E_i(x)$, where $\text{TopK}(G(x))$ denotes the set of indices of the top- k experts selected by the gating network for input x .

4 ForceVLA

Portable idea

FVLMoE block after VLM, before / into FM suffix.

CLEAR compatible

OpenPI π_0 checkpoints that retain FM + F/T datasets.

NOT measured

$\pi_{0.5}$ / KI / RTC numerical gains. Do not invent.

Rule

Force AFTER VLM. Early MoE $\rightarrow$ 0% moral.

What a BEHAVIOR contact stack must stream

6D wrench synced

Time-synced with RGB + proprio (ForceVLA-Data pattern). World/TCP frame consistent.

Latency vs chunk rate

Wrench must arrive in time for FM chunk conditioning. Watch delay as a future ablation (paper lacks it).

Integrated vs retrofit

Flexiv-class works in-paper. Authors flag lower-cost external F/T as accessibility future work.

Estimated wrench caveat

Paper limitation: estimated external wrench today. Prefer high-fid transducers when porting.

Tactile skins parallel

TLA / Tac-Man family — not ForceVLA results. Same MoE could host image+F/T tokens later (TENTATIVE).

No F/T = blocked

Without a force stream, this blueprint does not apply. Keep vision VLA for free-space language tasks.

Contact-rich VLAs need sync F/T demos — not OXE alone

A Data Collection System

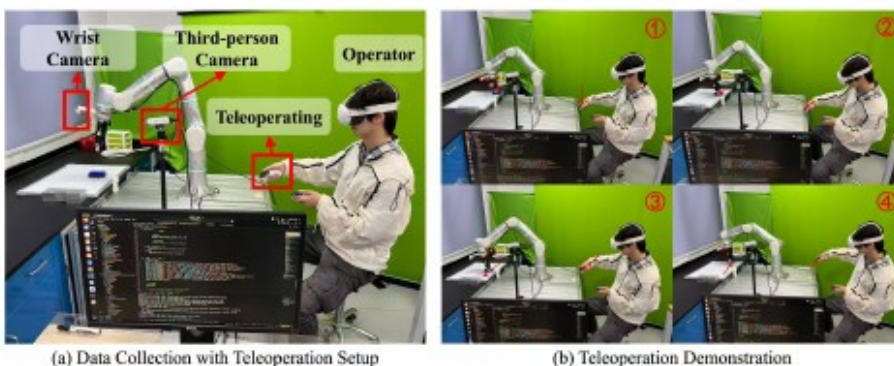


Figure 8: Our data collection system setup.

Our data collection system is depicted in Figure 8(a). The setup features a robotic arm equipped with both a wrist-mounted camera and a static third-person view camera to capture diverse visual perspectives. An operator, wearing a Quest 3 headset and using hand-held controllers, teleoperates the robot arm. A nearby computer runs the necessary software for data acquisition, which includes

ForceVLA-Data

244 traj

~140k steps

~50 demos / task

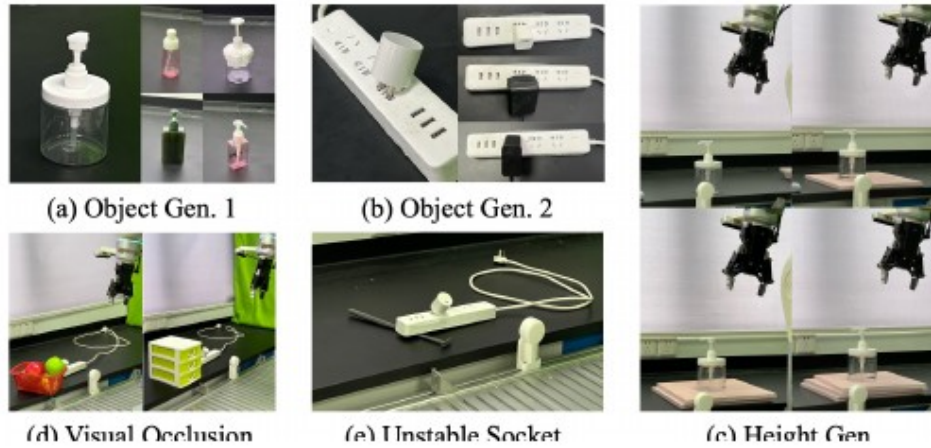
Implication

BEHAVIOR contact suites need wrench-synced teleop or replay — language-only corpora will not unlock FVLMoE.

Release

Code/data 'will be released' — pin when public.

When vision fails in clutter — force must carry



Visual Occlusion

90%

ForceVLA vs $\leq 60\%$ π_0 -base

Household read

Cable seating behind furniture; socket under clutter; wrist cam blocked by tool.

Still hard

Unstable Socket 20%. Physical instability $\neq$ solved by force MoE alone.

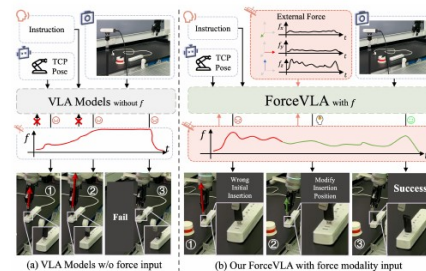


Figure 1: Comparison between ForceVLA and baselines without force input. Without force feedback, the policy fails to correct pose errors and completes insertion incorrectly. In contrast, ForceVLA leverages external force signals to adjust insertion strategies dynamically, leading to successful

Seven threads — none invent paper numbers

1 Port FVLMoE

Onto open π_0 / OpenPI with public F/T datasets. Drop-in late block.

2 High-fid wrench

Replace estimated wrench; ablate noise / delay (absent in paper).

3 Tactile + F/T MoE

Add tactile image tokens beside wrench in the same sparse MoE.

4 BEHAVIOR contact eval

Insert / wipe / open categories with forced occlusion. No invented SR here.

5 Expert load balance

Study Expert-0 generalist bias; rare contact events need capacity.

6 Low-cost retrofit F/T

Paper's stated accessibility direction on common arms.

7 USB-hard curriculum

Force alone insufficient for fine alignment (multi-task USB 10%).

— Refuse

Discord/Drive packs. Invented BEHAVIOR baseline. Invented $\pi_{0.5}$ bake-off.

Kitchen / clean / food / occluded seating

Kitchen plugs / appliances

Insert assist with force-corrected seating when sockets are occluded or soft.

Wipe / scrub profiles

Cleaning robots that maintain contact force, not just end-effector paths.

Food prep peel/cut

Force phases for peel/press. Continuous metrics matter (Tab.1 moral).

Occluded cable seating

Behind furniture / under desks — occlusion 90% is the analogy, not a BEHAVIOR score.

Torque-limited depth

Height Gen lesson: scale interaction forces; avoid torque limit violations.

What not to pitch

Force replacing language grounding. ForceVLA as $\pi_{0.5}$ replacement. Official BEHAVIOR'26 baseline.

Say the hedge out loud

TENTATIVE useful research recipe — NOT a ForceVLA paper claim:

1. Run a $\pi_{0.5}$ -style high-level semantic / open-world loop for language grounding.
2. Keep a low-level FM action expert for chunks.
3. When wrist F/T is present, augment with FVLMoE late force fusion for contact phases.
4. Eval contact categories under occlusion separately from free-space language success.

CLEAR: ForceVLA measured force MoE on π_0 FM only. No $\pi_{0.5}$ / RTC numbers exist to cite.

CLEAR: no Discord, no Drive, no invented BEHAVIOR baseline status.

2025 blueprint — not a $\pi_{0.5}$ replacement

For BEHAVIOR contact-rich homes, pair modern VLAs with

wrist F/T + late MoE fusion

so insertion / wipe / occlusion phases stop being pure vision failures.

ForceVLA is the 2025 blueprint, not a $\pi_{0.5}$ replacement.

Cite: 60.5 vs 37.3 (+23.2); early MoE 0%; occlusion 90%. Refuse invented BEHAVIOR SR.

LAST Alt row locked — force-in-VLA on the record

Stakeholder

Buy late MoE when F/T exists. Don't concat-only. FAST+naive hurts.

Reviewer

Late MoE real contrib. Soft USB 10% multi / unstable / no $\pi_{0.5}$.

Empiricist

Ablation Bot ranks. Smoke $\neq$ Fig.5. Pump honesty.

Archaeologist

$\pi_0 \rightarrow$ late force $\rightarrow$ FVLMoE $\rightarrow$ FM. CLEAR vs MoPA / JEPA.

Visionary

BEHAVIOR contact-rich + F/T + late MoE. NO Discord/Drive.

Shared spine

60.5 / 37.3 / +23.2 · early 0% · occlusion 90%
· not MoPA · not JEPA · no $\pi_{0.5}$ bake-off.